

Topic: Graphical Password authentication with segmentation and rumbling, recognition based authentication, recall based authentication, colour grade based authentication and click points recall authentication.

Introduction

In this modern age of increasing interconnectivity, information security has come to occupy a pivotal position in recent research works. Traditional methods of authentication have been text-based password schemes. However, the major drawback with them, has been the fact that their strength varies inversely with user convenience. Users generally tend to select passwords that are easier to recall and shorter in length. This, gives rise to password cracking attempts. A graphical password is a confirmation framework that works by having the client selected pictures, in a particular order.

The project allows user to input an image as an alternative to a textual password, the system segments the image into an array of images and stores them accordingly. The next time user logs on to the system the segmented image is presented by the system in a jumbled order. In order to login successfully, the user has to arrange the jumbled images. The project allows user to input an image as its password and only user knows what the image looks like as a whole. On receiving the image the system segments the image into an array of images and stores them accordingly. The next time user logs on to the system the segmented image is received by the system in a jumbled order. Now if user chooses the parts of image in an order so as to make the original image he sent then user is considered authentic. Else the user is not granted access. The system uses image segmentation based on coordinates. The coordinates of the segmented image allow the system to fragment the image and store it in different parts. Actually system segments the image into a grid and stores each part accordingly in order. But while logging in the image is shown as broken and in a jumbled order. At this time only the user who provided the image knows what the actual image looks like and he must the parts in horizontal direction from left to right one row at a time according to the order in which parts were arranged in the original image. So the user is granted access after successful attempt. if not it alerts the users through mail.

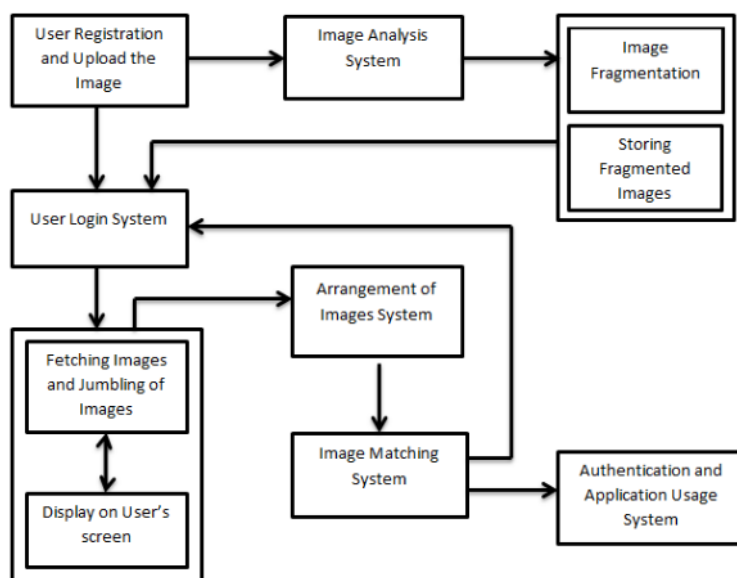
Existing System

Currently most online applications use text based authentication system and some demand combination of letters, digits and symbols. Traditional methods of authentication have been text-based password schemes. However, the major drawback with them has been the fact that their strength varies inversely with user convenience. Users generally tend to select passwords that are easier to recall and shorter in length. This, gives rise to password cracking attempts. A hacker can use key loggers, screen capturing apps to hack the passwords you are entering in online applications. Certain applications use face identification as an option for access authentication.

Proposed System

The project allows user to input an image as an alternative to a textual password, the system segments the image into an array of images and stores them accordingly. The next time user logs on to the system the segmented image is presented by the system in a jumbled order. In order to login successfully, the user has to arrange the jumbled images. The project allows user to input an image as its password and only user knows what the image looks like as a whole. On receiving the image the system segments the image into an array of images and stores them accordingly. The next time user logs on to the system the segmented image is received by the system in a jumbled order. Now if user chooses the parts of image in an order so as to make the original image he sent then user is considered authentic. Else the user is not granted access. The system uses image segmentation based on coordinates. The coordinates of the segmented image allow the system to fragment the image and store it in different parts. Actually system segments the image into a grid and stores each part accordingly in order. But while logging in the image is shown as broken and in a jumbled order. At this time only the user who provided the image knows what the actual image looks like and he must the parts in horizontal direction from left to right one row at a time according to the order in which parts were arranged in the original image. So the user is granted access after successful attempt. If the user fails to accomplish it in certain duration of time the login will be blocked temporarily and give login attempt alert to the mail provided during registration.

Architecture Diagram



. Graphical passwords authentication system has inherent advantages over conventional textual password schemes. In this system there are 3 authentication levels after registering the basic information, User login's and after implementing id password system ask to authenticate in 3 different levels.

Functional Modules

- **Image Submission Module:** This module collects data from the user along with an image to be used as a password and store it in the database.
- **Image Fragmentation Module:** This module selects a suitable algorithm for image fragmentation, applies the algorithm on it and prepares them for storage.
- **Image Fragments Storage Module:** This module checks for contiguous spaces in the memory and store the fragments in it.
- **Images Jumbling Module:** This module jumbles the images row-wise and displays them on the users screen so that he/she can arrange the images in correct order.
- **Authentication Module:** This module matches the arranged image with the original image and authenticates the user if they matches otherwise asks the user to try again.
- **Alerts module:** if the user fails to login within a certain span of time, the system will automatically send login attempt alerts to mail provided during registration.

Authentication types

Recognition based authentication: A user is given a set of images and he has to identify the images he selected during registration.

Recall based authentication: A user is asked to reproduce something that is created or selected at registration stage. For example in pass point scheme, a user can click any point at an image to create a password and tolerance around each pixel is calculated. During authentication user have to select the points within the tolerance in the correct sequences to login.

Cued Recall: Cued Click points are an alternative to pass points technique. In CCP user click one point of each image rather than on five points on one image. It offers cued recall and instantly alerts the users if they make a mistake while entering their latest click point

Colour grade based authentication(alternating session passwords): In this method certain colour gradients are randomly selected and certain values are automatically assigned to them. User will select those paired passwords from a randomly generated matrix using algorithms to get the authentication.

This web can be secured with alternating session passwords by combination of colors and texts/digits to overcome the vulnerability to eves dropping, dictionary attacks and shoulder surfing. During registration, user should rate colors as a password to login into system as shown in figure 1. The User should rate colors from 1 to 8 and he can remember it as “RLYOBGIP”. Same rating can be given to different colors. During the login phase, when the user enters his username an interface is displayed based on the colors selected by the user. The login interface consists of grid of size 8×8. This grid contains digits 1-8 placed randomly

in grid cells. The interface also contains strips of colors as shown in figure 2. The color grid consists of 4 pairs of colors. Each pair of color represents the row and the column of the grid.

1	5	6	3	2	7	4	8

Figure 1 : Rating of colors by the user

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	1	2	3	4	5	6	7	8
1	5	6	3	4	8	1	2	7
2	2	8	6	5	1	3	7	4
3	4	3	7	8	2	5	1	6
4	7	4	1	2	5	6	3	8
5	8	2	4	6	3	7	5	1
6	1	5	8	7	4	2	6	3
7	3	7	5	1	6	4	8	2
8	6	1	2	3	7	8	4	5

Figure 2: Login interface in payments

Figure 2 shows the login interface having the color grid and number grid of 8 x 8 having numbers 1 to 8 randomly placed in the grid. Depending on the ratings given to colors, we get the session password. As discussed above, the first color of every pair in color grid represents row and second represents column of the number grid. The number in the intersection of the row and column of the grid is part of the session password. Consider the figure 1 ratings and figure 2 login interface for demonstration. The first pair has red and yellow colors. The red color rating is 1 and yellow color rating is 3. So the first letter of session password is 1st row and 3rd column intersecting element i.e 3. The same method is followed for other pairs of colors. For figure 2 the password is “ 3573 ”. Instead of digits, alphabets can be used. For every login, both the number grid and the color grid get randomizes so the session password changes for every session.

HARDWARE REQUIREMENTS:

SERVER:

1. Processor : AMD or higher
2. Memory : 8 GB or more
3. Hard disc : 100 GB or more
4. Network card

CLIENT:

1. Processor : Any processor
2. Memory : 512 MB or more
3. Hard disc : 10 GB or more
4. Network card

SOFTWARE REQUIREMENTS:

1. Front-end : ,asp C#.NET, JavaScript with C#,
2. Back-end : Microsoft SQL Server 2018
3. OS : Windows XP/Vista/7/8/10
4. Server : IIS-7
5. Editor : Visual Studio 2020